

Chapter 3

System Development Approaches

Basically, there are 2 ways to implement SDLC. IT projects today will follow either a structured approach or a Rapid Application Development (RAD) which is a newer approach.

Structured Approach to System Development

The structured approach to system development has been around since 1960s and 1970s when large mainframe applications were developed

Applications then were built when:

- 1) Systems were fairly simple and independent from one another
- 2) Computer hardware was relatively more expensive than labour
- 3) Development and programming tools were primitive compared to today (Satzinger, Jackson & Burd, 2002).

The waterfall method in Fig 1.4 follows SDLC in a very sequential and structured manner e.g. design activities start only after analysis and requirements activities have been completed. Subsequently, development or programming activities would not commence until the design phase is complete. Even though one can go back to previous stages, it's not always easy or desirable.

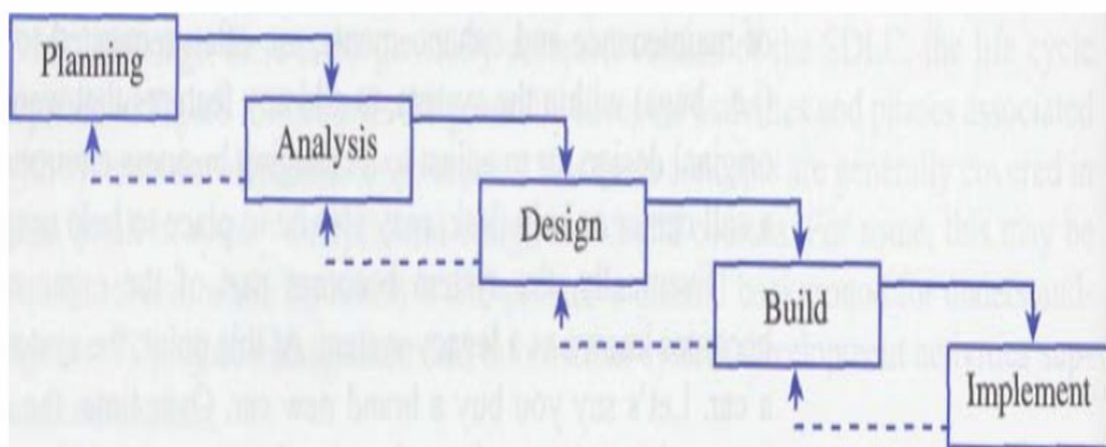


Figure 1.4 Waterfall Model

Source: Marchewka, 2003

Planning overhead is minimised because it's all done up-front and a tangible system is not produced until the end of life cycle (McConnell, 1996). The notion of waterfall is a metaphor of a number of activities from one phase to the next. This approach emphasises a sequential and logical flow of development activities.

This approach is suitable when developing structured systems plus assumes that requirements defined in analysis stage don't change significantly over the rest of the project. In addition because it will provide a firm structure that can minimise wasted effort this approach may work well when the project team is inexperienced or less technically competent (McConnell, 1996).

Rapid Applications Development (RAD)

On the other hand, one can take a less-structured approach to designing systems. Today, taking less time to develop and implement an IS can provide organisation with competitive advantage. Furthermore as evidenced by CHAOS study, larger projects that take longer to develop are riskier than smaller and shorter projects.

Satzinger, Jackson & Burd (2002) defined RAD as “a collection of development approaches, techniques, tools and technologies, each of which has been proven to shorten development schedules under some conditions.” This implies that various development approaches, tools and techniques and so on can be mixed and matched based on the project. For some projects, it implies that waterfall approach is most appropriate. Nonetheless, RAD often follows one of the following iterative approaches:

Prototyping

Prototyping is an iterative approach to systems development. Here the developer and user work together very close to develop a partially or fully functional system as soon as possible probably within a few weeks. Prototyping will go through a number of iterations as functional requirements are defined and changed.

This approach is most helpful when the requirements of new system are difficult to define or when working with a new technology where the capabilities of the technology are unknown or not very well understood. A prototype may be either a throwaway system or a fully usable system.

Throwaway prototype could be used to discover or refine requirements specifications that can be used as a model for developing the real system. On the other hand, a prototype may become the actual system after going through a number of refinements over time.

Spiral Development

Another way to expedite SDLC is the spiral approach proposed by Boehm (1998). The spiral model offers a risk-oriented approach in which software project is split into a number of mini projects where each addresses one or more risks until all major risks have been addressed.

A risk can be defined as a poorly understood requirement or architecture or a potential problem with technology or system performance. The basic idea in developing a system is to start on a small scale where risk can be identified. When identified, the development team then develops a plan which addresses these risks and evaluates various alternatives. Deliverables for iteration are identified, developed plus verified before planning and embarking on the next iteration.

Thus completing each iteration brings the project closer to a fully functional system. Each iteration should be reviewed as it offers a way of controlling project's overall risk. The main challenges will crop up early in project thereby providing the chance to reduce the total cost of project. The disadvantage of spiral development approach centres on its complexities (Satzinger, Jackson &

Burd). Complexities in managing projects because many people are working on a number of parallel activities.

Extreme Programming (XP)

Kent Beck introduced idea of XP in the mid 1990s. Under XP, system is transferred to users in a series of versions called releases. A release would be developed by deploying several iterations developed and tested within weeks or months. Any release is a working system that only includes one or several functions which are part of a complete system specifications.

The XP embodies a number of activities where user requirements are documented first as a user story. Object-oriented model known as class diagram is used to document user stories. So a release is developed over the course of several iterations. A set of acceptance tests for each story is then developed. The Releases that pass acceptance test are then considered complete The XP offers continuous testing and integration of different software modules and components while supporting active user involvement.

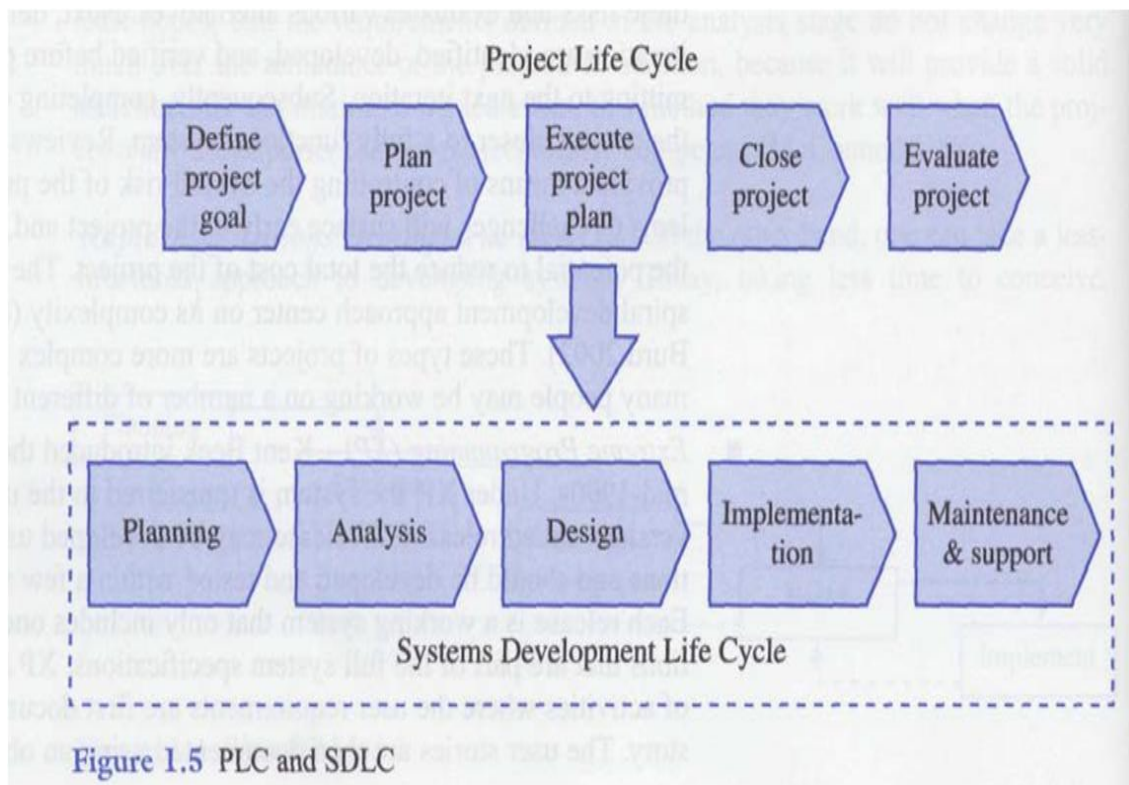
In addition, XP often incorporates team programming where 2 programmers work together on the same workstation (WS). Small teams of developers often work in a common room where WSs are positioned in the centre and workspace for each team member is positioned around the perimeter. Developer often, are prohibited from working more than 40 hours a week in order to avoid burntout and mistakes that often occur because of fatigue (Satzinger, Jackson & Burd).

PLC versus SDLC 1/5

You may still be wondering about the difference between PLC and SDLC. Even though they may seem to be similar, the difference is that PLC focuses on processes of managing a project while SDLC focuses on creating and implementing a product i.e. the IS. But in this context, our focus is solely on PLC

Although SDLC and the particular approach chosen will have a direct impact on a project's scope (i.e. deliverables that project team must provide) including work activities needed to produce those deliverables, in effect, activities number, their sequence, time to complete and resources required will directly establish the project's schedule and budget.

As depicted in Fig 1.5, SDLC is really part of PLC since most of the activities for developing IS take place during the execution phase.



While the last 2 stages of the PLC – closing and evaluating project take place after IS implementation, integration of PM and system development activities is one vital component that differentiates IT projects from other types of projects.

Project management Knowledge Areas

The Guide to PMBOK defines 9 knowledge management areas for understanding project management. These 9 knowledge management areas are portrayed in Fig 1.6 (PMBOK).

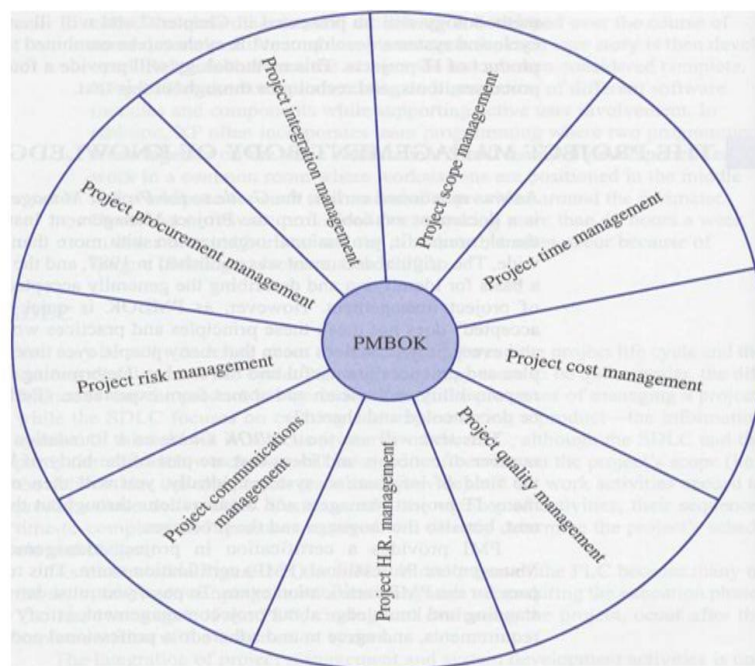


Figure 1.6 Project Management Body of Knowledge (PMBOK)

Project Integration Management

Integration focuses on coordinating project's plan development, execution and control of changes.

Project Scope Management

Project scope is work to be completed by project team. Scope management offers assurance that project's work is defined accurately and completely plus that it is completed as planned. Furthermore, scope management embodies ways to ensure that proper scope changes procedures are in place.

Project Time Management

Time management is vital for developing, monitoring and managing projects' scope. This includes identifying project's phases and activities plus estimating, sequencing and designing resources for each activity to ensure that the project's scope and objectives are met.

Project Cost Management

Cost management makes sure that project's budget is developed plus completed as approved.

Project Quality Management

Quality management focuses on planning, developing and managing quality environment which allows the project to meet or exceed stakeholders' needs or expectations.

Project Human Resource Management

People are the most important resource on a project. Human resources management focuses on creating and developing a team as well as understanding and responding appropriately to project's stakeholders.

Project Communications Management

Communications management entails communicating timely and accurately information about the project to project's stakeholders.

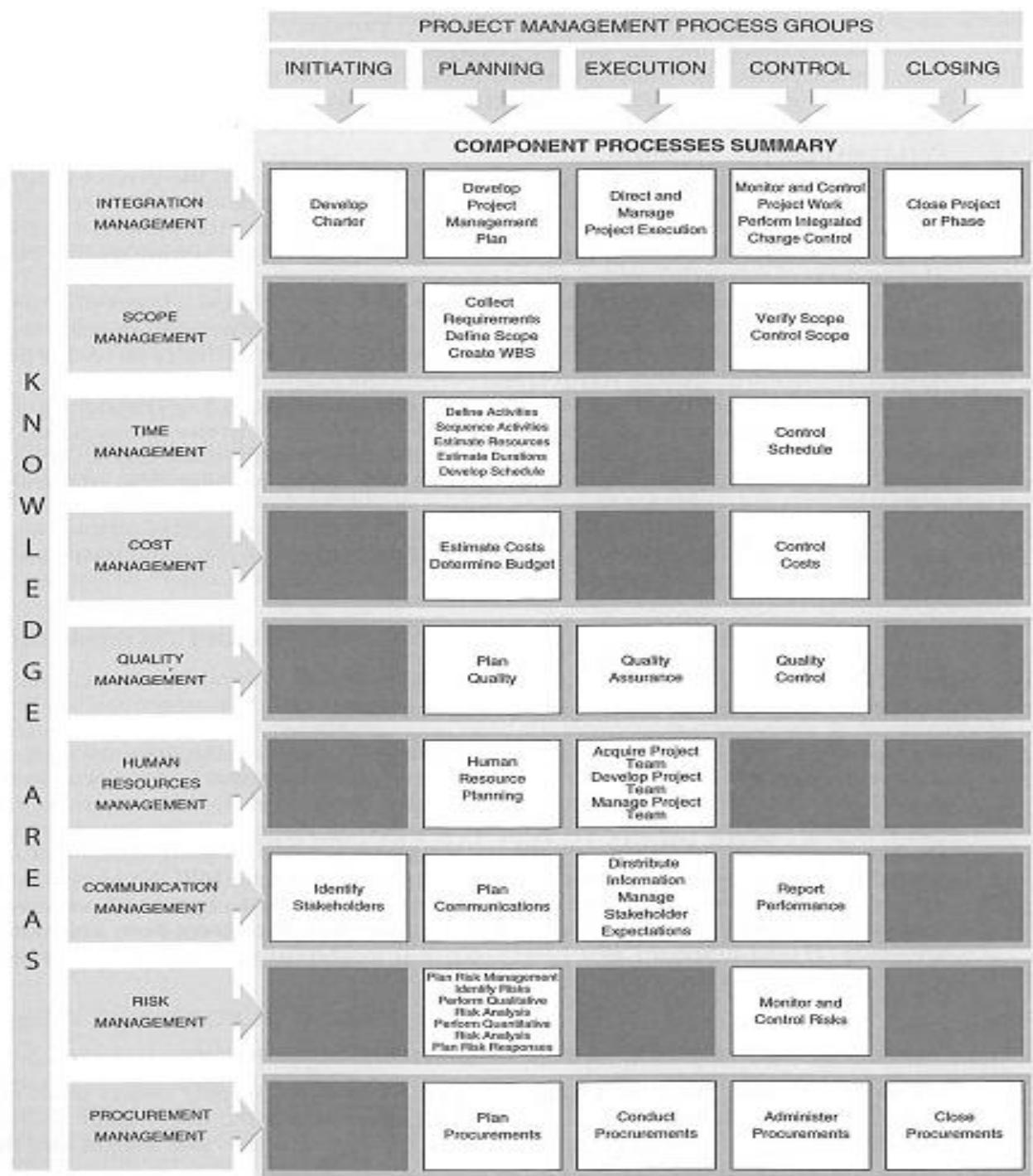
Project Risk Management

Every project faces a certain amount of risk. Project risk management is concerned with identifying and responding appropriately to risks that can impact the project.

Project Procurement Management

Projects time and time again need resources (people, hardware, software, etc) which are outside organisation. So procurement management makes certain that these resources are acquired properly.

Summary of Knowledge Areas (Brewer & Dittman, 2010)



Introduction to Project Selection and Initiation

Before initiating the start of a project, project managers need to take a step back plus figure out what project the organisation should embark on. In the past, many projects have been selected because of a “squeaky wheel” syndrome i.e. the person who makes the most noise repeatedly gets his or her project selected. You can see that this is not the best approach to choosing a project.

An organisation needs to go through a formal process of selecting the right mix of projects before formalising list into a project charter plus start work. In most organisations, there are more projects

that there are funds plus people to run them. Organisations would need to be very selective when choosing projects and these should be the ones that will return most value to the organisation.

Value in this case does not only relate to dollars (profit) but factors such as market share, customer relations, safety issues, environmental concerns and new government regulations

Questions like!

- Why this project?
- Who selected it?
- And what else is going on in the organisation at the same time?

These are all questions that require being asking plus answering before work on project commences. You only need to look at the dismal success rate of IT projects to be aware of why it's vital that organisations begin by selecting the right projects to work on. In the past, IT professionals have carried out a poor job of understanding and presenting the value of IT project

The reasons for this poor performance are two folds:

First – IT professionals spend more time studying technology than finance

Second – they are unable to communicate effectively to management that IT projects are different from projects in other industries and therefore require different metrics. IT professionals most specifically managers and project managers must have an understanding of the broader picture of the organisation plus impact that a technology project will have on it from cost and other perspectives. They should understand and possess ability to communicate the value of IT projects in terms that the rest of organisation can understand.

We will describe some of the techniques that will help project manager to communicate the value of IT projects. In today's business climate of tight budgets, limited IT resources and low IT projects, organisations are becoming more and more selective in IT projects they approve. Also most organisations are starting to realise the strategic importance of IT to their success. A survey commissioned by InfoWorld in 2002 found out that 70% of respondents said that IT was "absolutely essential to their company's business objectives" (InfoWorld, 2002). To get projects approved, IT managers must learn to align their projects with the company's strategic direction. To achieve this, project managers must ask themselves some key questions concerning a project so that they can understand what value it provides to the organisation, the risks associated with the project and project value to organisation if it's successful. Andriole (2001) identified 4 key issues that must be considered in order to understand the bigger organisational picture in which IT projects play a role. They include: business value, technology, cost/benefit questions and risks.

Business Value

IT managers must look at a project from a business perspective through identification of what business process(es) will be affected. They need to thoroughly understand what impact a project will have on these processes. They must also understand what impact a project will have on associated processes. Deploying systems approach assists in learning what impact a project will have on all parts of the organisation.

Technology

Technology utilise for a project needs to be well tested, scalable, secure, modifiable and usable. There are some questions that need to be asked:

- Has technology been deployed in the organisation before and do we have experience with it?
- What technology is the competition using?
- And how might technology permit the organisation to see what works?
- And what does not work in a specific industry?

Cost/Benefit Questions

Organisations need to understand whether the complete cost including acquisition, development and ongoing support costs of projects outweigh its benefits.

Risks

IT project managers must do a thorough risk assessment for the project. They must know what kinds of issues or problems that might crop during the project such as software vendor going out of business in the middle of the project or funding for project being cut in half during execution plus be sure to have appropriate safeguards or workarounds in place so that the project can still be completed. All of this information needs to be presented in the project's business case.